

Here are some commonly used options in **tcpdump** and their working:

1. -i <interface>

- **Description:** Specifies the network interface to listen on.
- **Example:** `tcpdump -i eth0`
- **Working:** Captures packets on the eth0 network interface.

2. -c <count>

- **Description:** Specifies the number of packets to capture.
- **Example:** `tcpdump -c 10`
- **Working:** Captures 10 packets and then stops.

3. -v

- **Description:** Increases the verbosity of the output.
- **Example:** `tcpdump -v`
- **Working:** Provides more detailed information about each packet.

4. -vv

- **Description:** Increases verbosity further.
- **Example:** `tcpdump -vv`
- **Working:** Shows even more detailed information, such as TTL, length, etc.

5. -vvv

- **Description:** Provides maximum verbosity.
- **Example:** `tcpdump -vvv`
- **Working:** Displays full details, including packet data, flags, and other information.

6. -n

- **Description:** Disables hostname resolution. IP addresses will be shown instead of domain names.
- **Example:** `tcpdump -n`
- **Working:** Displays raw IP addresses instead of trying to resolve them into hostnames.

7. -nn

- **Description:** Disables both hostname and port number resolution.
- **Example:** `tcpdump -nn`
- **Working:** Displays IP addresses and port numbers in numeric form, avoiding both hostname and port name resolution.

8. -X

- **Description:** Shows the packet contents in both hexadecimal and ASCII.
- **Example:** `tcpdump -X`
- **Working:** Displays the packet data in hex and ASCII format, making it easier to analyze raw data.

9. -e

- **Description:** Displays the link-layer header (Ethernet headers, for example).
- **Example:** `tcpdump -e`
- **Working:** Shows Ethernet header information, such as MAC addresses and packet type.

10. -w <file>

- **Description:** Writes the captured packets to a file in pcap format.
- **Example:** `tcpdump -w capture.pcap`
- **Working:** Saves the captured packets into the specified file, which can be later analyzed with Wireshark or other tools.

11. -r <file>

- **Description:** Reads packets from a previously saved pcap file.
- **Example:** `tcpdump -r capture.pcap`
- **Working:** Reads and displays the packets stored in the specified pcap file.

12. -s <snaplen>

- **Description:** Defines the snapshot length (how much of each packet to capture).
- **Example:** `tcpdump -s 1024`
- **Working:** Captures up to 1024 bytes of each packet. The default is usually 68 or 96 bytes, which may truncate large packets.

13. -t

- **Description:** Disables the printing of the timestamp on each line.
- **Example:** `tcpdump -t`
- **Working:** Prevents the output from displaying the timestamp for each captured packet.

14. -l

- **Description:** Makes tcpdump write its output to stdout line by line.
- **Example:** `tcpdump -l`
- **Working:** Allows you to see packet data in real-time as it is captured, which can be useful for piping output to other programs.

15. -q

- **Description:** Reduces the verbosity of output.
- **Example:** `tcpdump -q`
- **Working:** Shows fewer details, just the most essential packet information like IP addresses and ports.

16. -A

- **Description:** Prints each packet's contents in ASCII.
- **Example:** `tcpdump -A`
- **Working:** Displays the payload data of the captured packets in ASCII, which can be helpful for viewing HTTP requests and responses.

17. -C <size>

- **Description:** Specifies the maximum file size (in MB) before creating a new file.
- **Example:** `tcpdump -C 10 -w capture.pcap`
- **Working:** When the capture file reaches 10 MB, a new file will be created to store further packets.

18. -G <seconds>

- **Description:** Rotates the capture file every specified number of seconds.
- **Example:** `tcpdump -G 60 -w capture-%Y-%m-%d_%H-%M-%S.pcap`
- **Working:** Creates a new capture file every 60 seconds and names it based on the timestamp.

19. host <hostname or IP>

- **Description:** Captures traffic to or from a specific host.
- **Example:** `tcpdump host 192.168.1.1`
- **Working:** Filters packets related to the host 192.168.1.1 (both incoming and outgoing).

20. port <port number>

- **Description:** Captures traffic on a specific port.
- **Example:** `tcpdump port 80`
- **Working:** Filters packets that are sent or received on port 80 (HTTP traffic).

21. src <IP>

- **Description:** Filters traffic from a specific source IP.
- **Example:** `tcpdump src 192.168.1.1`
- **Working:** Captures traffic coming from the IP 192.168.1.1.

22. dst <IP>

- **Description:** Filters traffic going to a specific destination IP.

- **Example:** `tcpdump dst 192.168.1.1`
- **Working:** Captures traffic destined for the IP 192.168.1.1.

23. icmp

- **Description:** Captures only ICMP traffic (ping requests, etc.).
- **Example:** `tcpdump icmp`
- **Working:** Filters packets based on the ICMP protocol, which is used for network diagnostics like pinging.

24. and, or, not

- **Description:** Logical operators to combine multiple filters.
 - **Example:** `tcpdump src 192.168.1.1 and dst port 80`
 - **Working:** Captures packets coming from 192.168.1.1 and going to port 80.
-

Example Combined Command:

- **`tcpdump -i eth0 -c 10 -n -w capture.pcap`**
 - Captures 10 packets from the eth0 interface.
 - Displays the output without resolving hostnames (-n).
 - Writes the packets to a file capture.pcap (-w).
-

Similar MCQs on tcpdump:

1. **Which option is used in tcpdump to display link-layer information?**
 - A. -x
 - B. -e
 - C. -n
 - D. -c
 - **Answer:** B. -e
2. **What does the -A option do in tcpdump?**
 - A. Prints packets in hexadecimal
 - B. Prints ASCII of the packets
 - C. Saves packets to a file
 - D. Displays the packet timestamp
 - **Answer:** B. Prints ASCII of the packets

3. Which tcpdump option prevents the display of timestamps in the output?

- ☐ A. -t
- ☐ B. -v
- ☐ C. -l
- ☐ D. -q
- ☐ **Answer: A. -t**

4. To capture traffic from a specific host, which tcpdump filter would you use?

- ☐ A. host <hostname or IP>
- ☐ B. port <port number>
- ☐ C. src <IP>
- ☐ D. dst <IP>
- ☐ **Answer: A. host <hostname or IP>**

5. Which tcpdump option is used to limit the capture to a specific number of packets?

- ☐ A. -c <count>
- ☐ B. -i <interface>
- ☐ C. -n
- ☐ D. -w <file>
- ☐ **Answer: A. -c <count>**